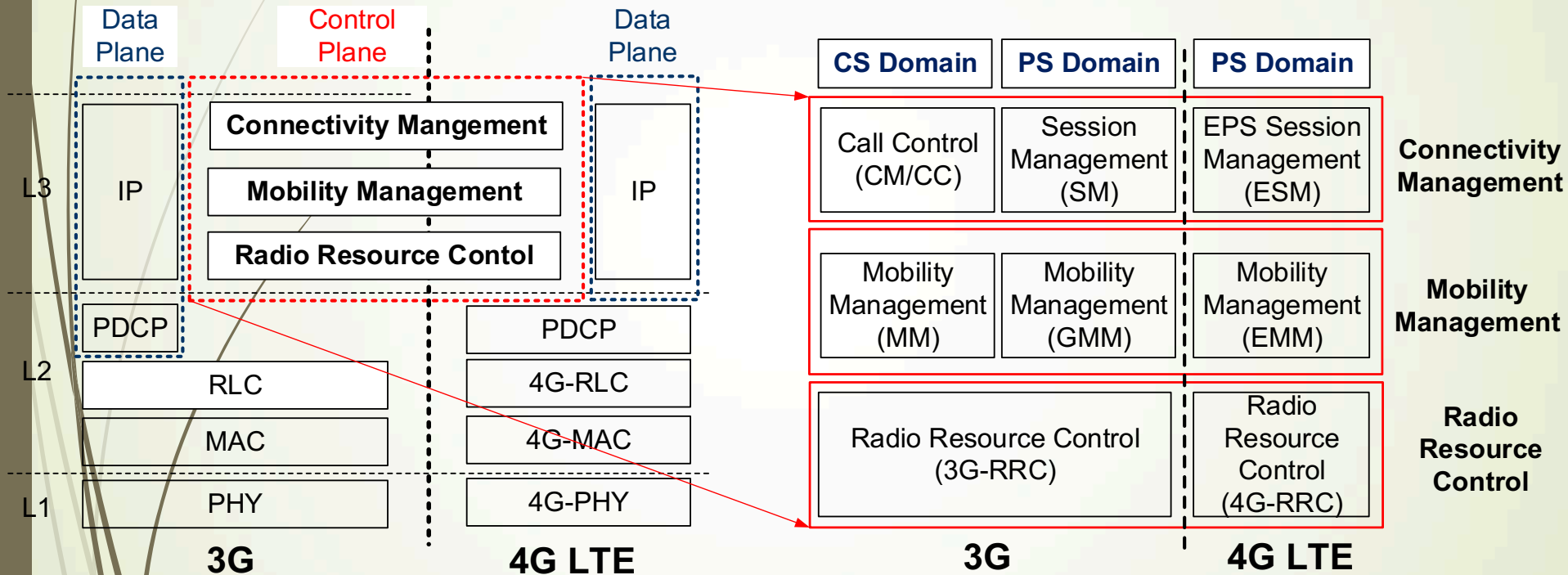


Layering Structure of 3G/4G Networks



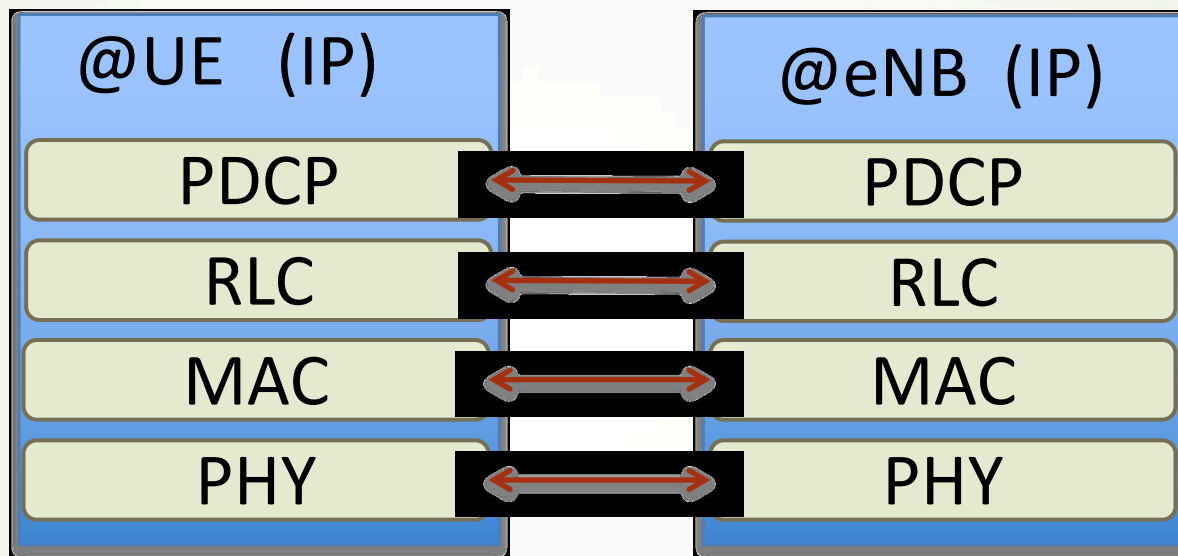
EPS: Evolved Packet System

PDCP: Packet Data Convergence Protocol

RLC: Radio Link Control

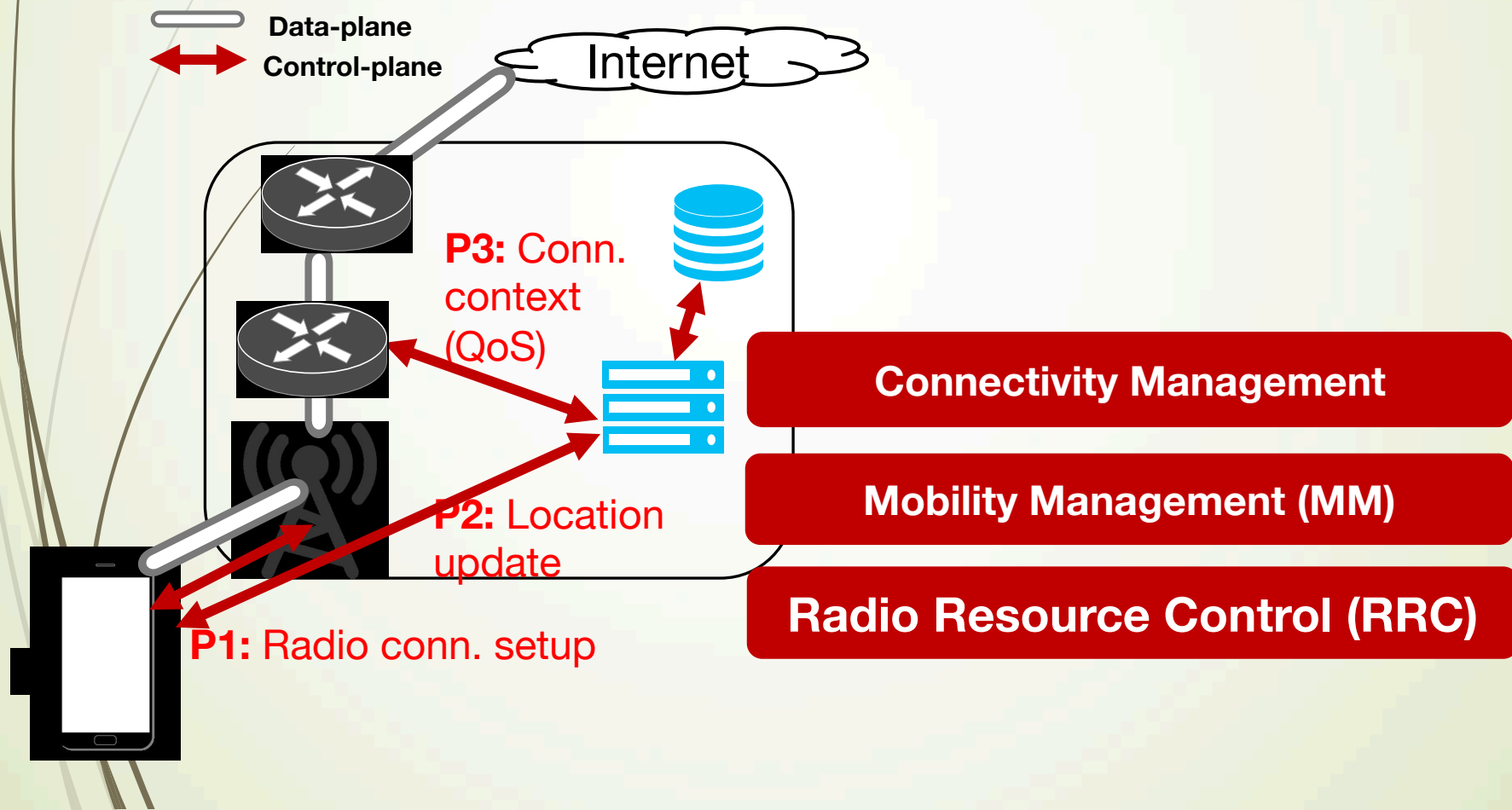
MAC: Medium Access Control

Data-plane Protocols



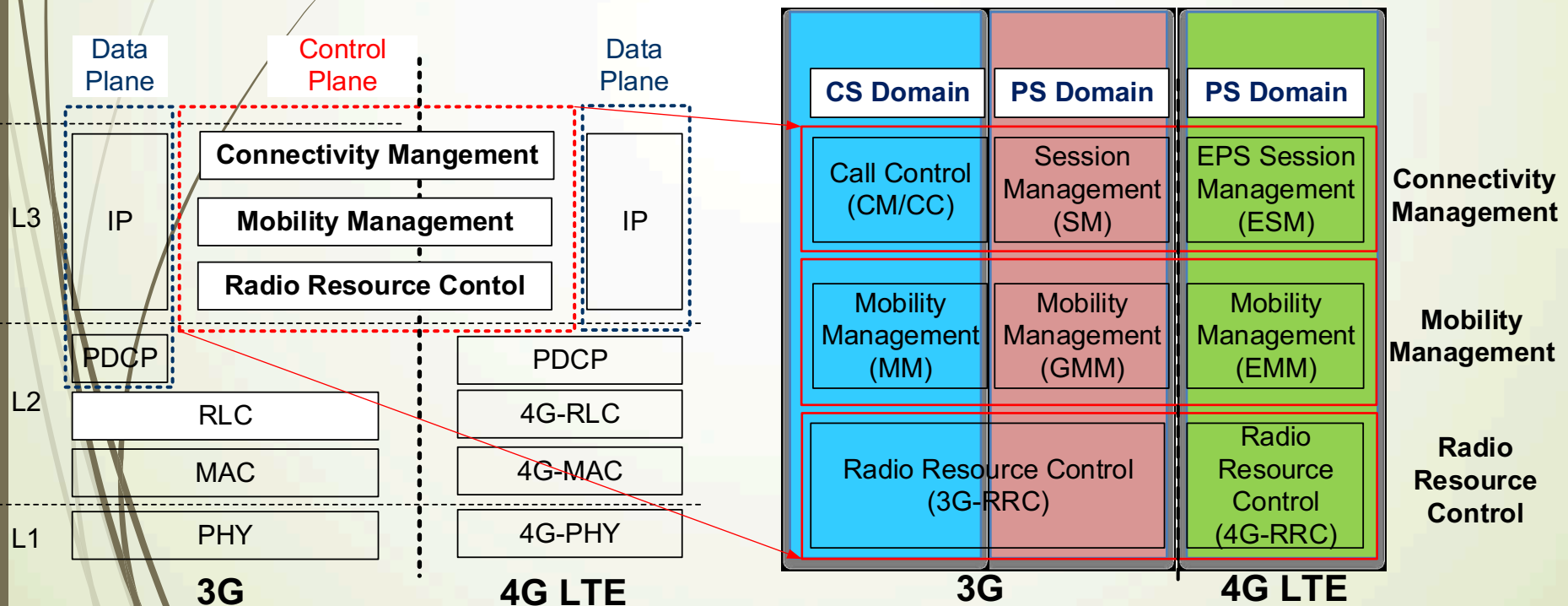
- **Packet Data Convergence Protocol (PDCP)** – header compression, radio encryption
- **Radio Link Control (RLC)** – Readies packets to be transferred over the air interface
- **Medium Access Control (MAC)** – Multiplexing, QoS

Jobs of Control-plane Protocols

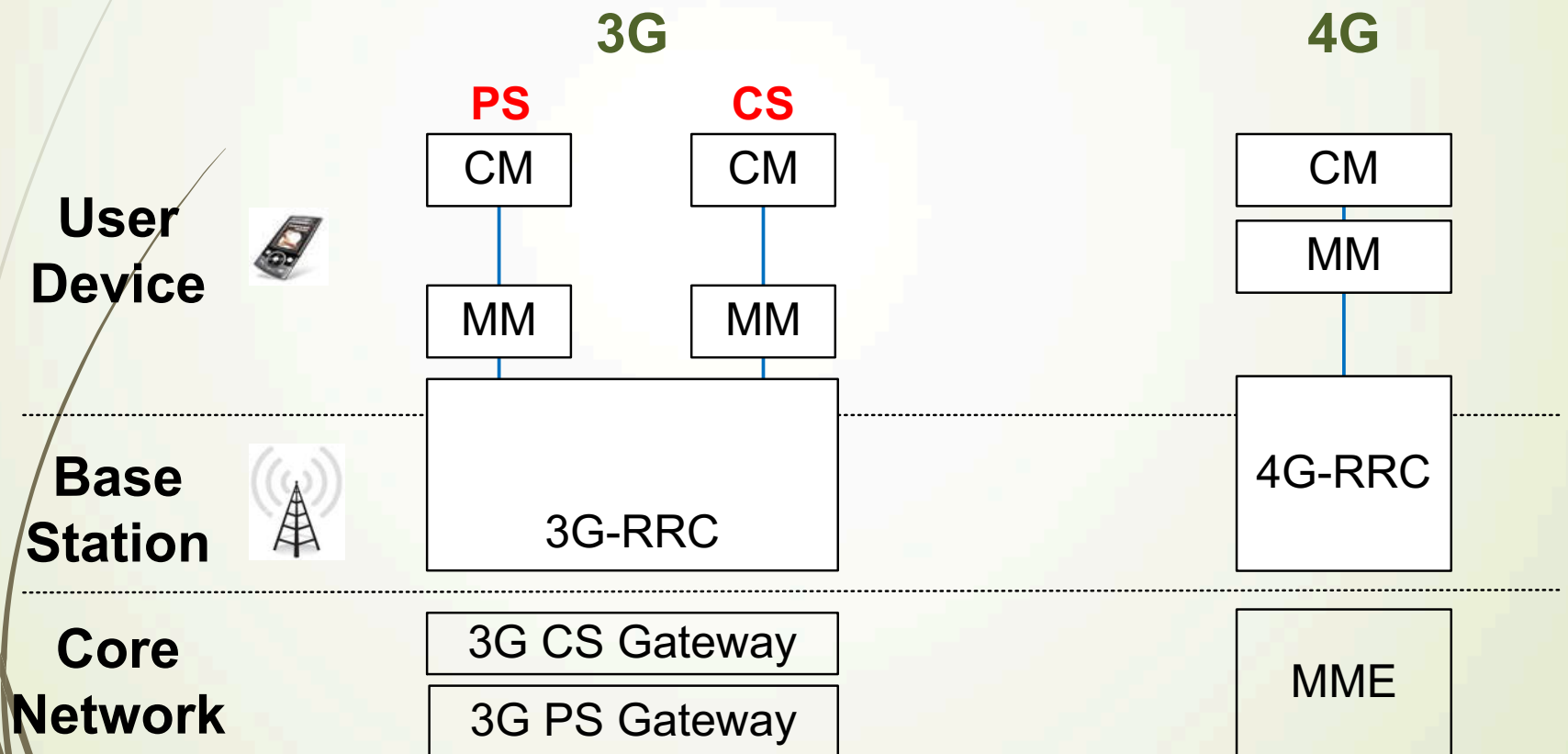


Control-Plane Protocols

- Variants of control functions
 - Hybrid 4G/3G systems
 - Voice versus data
 - Circuit- versus packet-switched

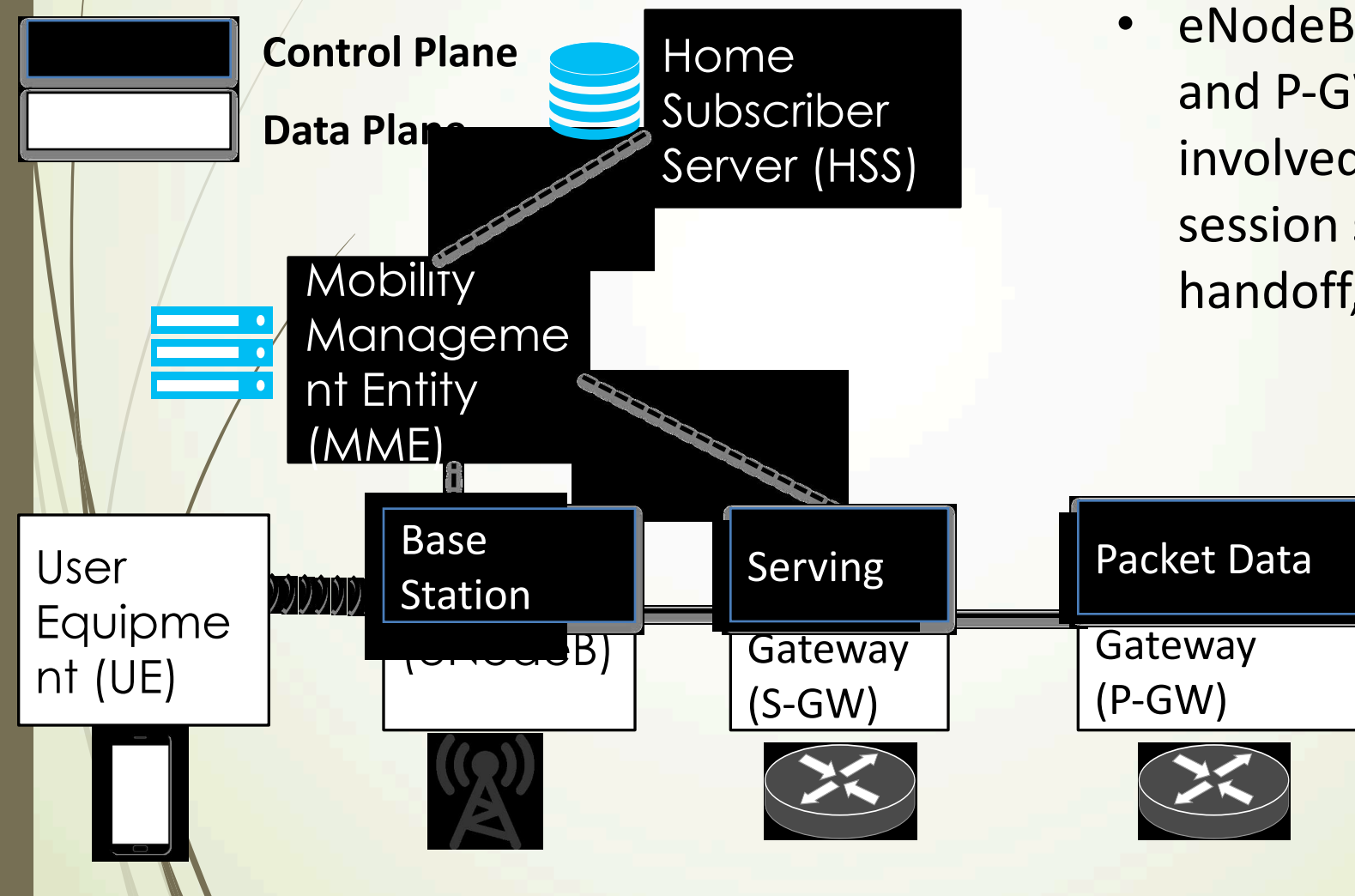


Devices, Base Stations, Core Networks



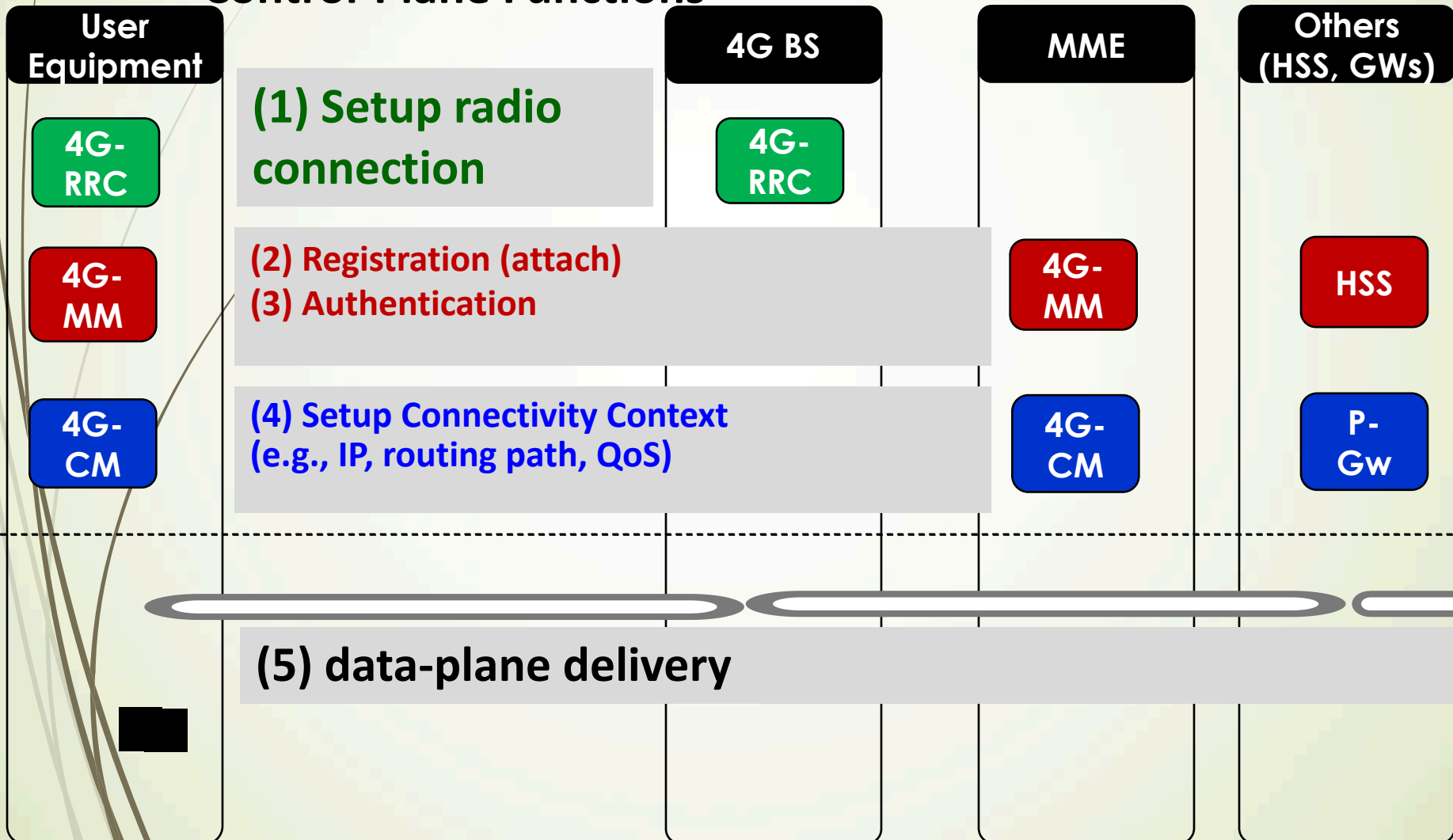
Control/Data Planes in LTE

- eNodeB, S-GW and P-GW are involved in session setup, handoff, routing



Putting Everything Together in LTE

Control-Plane Functions



Q: What is the largest difference between telecom networks and IP networks?

Questions

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